Hyperparathyroidism 1 (MONDO:0007767; OMIM 145000) — Comprehensive Disease Characteristics Report

Causal gene: *CDC73* (parafibromin), chromosome 1q31.2. Hyperparathyroidism 1 is caused by germline loss-of-function mutations of **CDC73** (formerly HRPT2 / C1orf28), the gene encoding the nuclear tumor-suppressor **parafibromin** (HGNC:16783; NCBI Gene 79577; UniProt Q6P1J9; Ensembl ENSG00000134371). Although the disease name is a serial number that does not name its gene, every identifier in this report follows MONDO:0007767 / OMIM 145000 / CDC73. This is the CDC73-related hereditary hyperparathyroidism spectrum, which in its fully expressed form is the **hyperparathyroidism–jaw tumor (HPT-JT) syndrome** and in its limited form is **familial isolated hyperparathyroidism (FIHP)**; both predispose to parathyroid carcinoma. It must not be confused with other primary-hyperparathyroidism loci such as MEN1 (MEN1 gene, 11q13), CASR-related familial hypocalciuric hypercalcemia, CDKN1B (MEN4), RET (MEN2A), or GCM2-related FIHP.

Summary

Hyperparathyroidism 1 is an autosomal dominant hereditary endocrine tumor-predisposition disorder arising from germline inactivating mutations in **CDC73** on chromosome **1q31.2**. CDC73 encodes **parafibromin**, a ubiquitously expressed 531-amino-acid nuclear protein that is a core subunit of the human **PAF1 complex (PAF1C)**, which controls RNA polymerase II transcriptional elongation and associated histone modifications. Parafibromin acts as a tumor suppressor principally by repressing the proliferation genes *CCND1* (cyclin D1) and *MYC* (c-myc), and it also participates in canonical Wnt/β-catenin signaling by binding directly to the C-terminal region of β-catenin. Disease results from **biallelic inactivation** of CDC73 in parathyroid chief cells — a germline "first hit" plus a somatic "second hit," most often loss of heterozygosity (LOH) on chromosome 1q — consistent with Knudson's two-hit model of hereditary cancer (PMID: 30799315).

Clinically, the disorder presents as **primary hyperparathyroidism (PHPT)** with markedly elevated PTH and often severe hypercalcemia, typically in young adults (mean age of presentation ~27–30 years, documented as early as the second decade). The syndromic form adds **ossifying (cemento-ossifying) fibromas of the mandible and maxilla**, **renal lesions** (cysts, hamartomas, Wilms tumor, mixed epithelial and stromal tumor), and **uterine tumors** (leiomyomas and adenofibromas). The feature that most distinguishes this syndrome from all other causes of PHPT is its **uniquely high lifetime risk of parathyroid carcinoma** — on the order of 15–37.5% in HPT-JT operative series, far above the <1% rate in sporadic PHPT. Risk is further stratified by genotype: **"high-impact" mutations that disrupt the parafibromin C-terminal domain confer ~6.6-fold higher carcinoma risk** than low-impact mutations (PMID: 32590342).

At the molecular level, loss of nuclear parafibromin (detectable by immunohistochemistry) is both a surrogate marker for CDC73 mutation and a predictor of malignant behavior. Somatic CDC73 mutations are found in up to 80% of sporadic parathyroid carcinomas, and ~30% of parathyroid carcinoma patients carry a germline CDC73 mutation, whereas benign sporadic parathyroid adenomas rarely harbor CDC73 mutations (PMID: 37531615; PMID: 16487440). A Cdc73-knockout mouse recapitulates the syndrome by developing parathyroid and uterine tumors, providing a validated in vivo model (PMID: 28288139). The cornerstone of management is **surgery** (en bloc resection when carcinoma is suspected), with lifelong surveillance of mutation carriers and cascade genetic testing of relatives.

Key Findings

Finding 1 — CDC73/parafibromin is the causal tumor suppressor, acting via a two-hit mechanism

Germline inactivating mutations of CDC73 (formerly HRPT2; encoding the ubiquitously expressed 531-amino-acid nuclear protein parafibromin) cause autosomal dominant HPT-JT, FIHP, and predispose to sporadic parathyroid carcinoma. As stated in PMID: 30799315: *"Loss of function of the cell division cycle protein 73 homolog (CDC73) gene is responsible for the syndrome. This gene encodes an ubiquitously expressed 531 amino acid protein, parafibromin, that acts as a tumor suppressor."* The somatic second hit is most often LOH at the CDC73 locus: *"Loss of heterozygosity (LOH) of the CDC73 locus in many HPT-JT associated parathyroid tumors*

from patients with germline mutation is in accordance with Knudson's 'two-hit' model for hereditary cancer." Loss of parafibromin by IHC serves as a surrogate marker (e.g., ~90% protein reduction documented in a carcinoma despite normal mRNA).

Finding 2 — Parafibromin functions in the PAF1 complex and regulates Wnt/β-catenin signaling and proliferation genes

Parafibromin is a core component of the human PAF1 complex controlling RNA Pol II transcriptional elongation and histone modifications. It represses *CCND1* (cyclin D1) and *MYC* (c-myc), and binds directly to the C-terminal region of β-catenin/Armadillo to transduce Wnt/Wg signals (Drosophila ortholog Hyrax/hyx). Per PMID: 29142233: *"CDC73/parafibromin was initially recognized as a tumor suppressor by inhibiting cell proliferation via repression of cyclin D1 and c-myc genes. In recent years, it has also shown oncogenic features by activating the canonical Wnt/β-catenin signal pathway."* And PMID: 16630820: *"Drosophila Hyrax and its human ortholog, Parafibromin, components of the Polymerase-Associated Factor 1 (PAF1) complex, are required for nuclear transduction of the Wnt/Wg signal and bind directly to the C-terminal region of beta-catenin/Armadillo."* Pathogenic missense mutations cluster in the N-terminal domain (residues 1–111; crystal structure solved to 1.02 Å) and disrupt its hydrophobic core/folding. Parafibromin additionally regulates the translational factor CPEB (PMID: 20339377).

Finding 3 — CDC73 genotype predicts parathyroid carcinoma risk (6.6-fold for high-impact/CTD mutations)

In a cohort of 419 CDC73 germline-mutation carriers, "high-impact" mutations (predicted significant conformational disruption or loss of parafibromin expression) were enriched among parathyroid-cancer cases. Per PMID: 32590342: *"The Kaplan-Meier analysis showed that high-impact mutations were associated with a 6.6-fold higher risk of parathyroid carcinoma compared to low-impact mutations, despite a similar risk of developing primary hyperparathyroidism between two groups."* Critically, *"Disruption of the C-terminal domain (CTD) of parafibromin is directly involved in predisposition to parathyroid carcinoma, since only the mutations impacting this domain were associated with an increased risk of parathyroid carcinoma."* Operative series report carcinoma in ~20–37.5% of HPT-JT PHPT patients; in one cohort (PMID: 25444225): *"37.5% of the patients developed parathyroid carcinoma (median overall survival, 8.9 years; median follow-up, 7.4 years)."*

Finding 4 — Clinical phenotype spectrum: PHPT with severe hypercalcemia, jaw fibromas, renal and uterine lesions

Core features are PHPT with markedly elevated PTH and severe hypercalcemia, ossifying fibromas of the mandible/maxilla, renal lesions (cysts, hamartomas, Wilms tumor, mixed epithelial/stromal tumor), and uterine tumors. PMID: 25959515 defines the syndrome as *"characterized by synchronous or metachronous occurrence of primary hyperparathyroidism (PHPT), ossifying fibroma of the maxilla and/or mandible, renal tumor and uterine tumors."* Onset is typically young adult (case-series means 27–30 years; as early as age 13). A 24-case jaw ossifying-fibroma review reported (PMID: 33505612): *"The average age was 28.68 years old (age range 10-66), with 12 male and 12 female patients (1:1 M/F ratio). Hyperparathyroidism results from parathyroid adenoma in 16/24 cases (66.666%) and parathyroid carcinome in 5/24 (20.833%)."* Penetrance of PHPT is age-dependent and incomplete — PMID: 33999366: *"Thirteen carried the deletion, the penetrance was estimated at 50% at 40 years."* PHPT is often uniglandular but multiglandular in ~31% and recurrent in ~20% (PMID: 25444225). Bone disease includes osteitis fibrosa cystica, brown tumors, and fractures.

Finding 5 — CDC73 maps to chromosome 1q31.2

CDC73 is located on chromosome **1q31.2** and contains 17 exons encoding the 531-amino-acid parafibromin (this corrects an earlier internal cytoband typo of "17q11.2"). The somatic second hit is commonly LOH on chromosome 1q. The locus is confirmed across multiple sources including the original mapping paper (PMID: 16487440) and the mouse-model study (PMID: 28288139).

Finding 6 — Somatic CDC73 in up to 80% of parathyroid carcinomas; germline in ~30%; benign adenomas rarely affected

Parathyroid carcinoma (PC) accounts for <1% of PHPT and ~0.005% of all cancers. Per PMID: 37531615: *"PC most commonly occurs as a sporadic disease and somatic CDC73 mutations can be detected in up to 80% of cases. Approximately 30% of patients harbor a germline mutation of the CDC73 gene."* In contrast, sporadic benign adenomas rarely carry CDC73 mutations. The original mapping (PMID: 16487440) states: *"The HPT-JT syndrome and FIHP are autosomal dominant disorders that may be caused by abnormalities of the HRPT2 gene, located on chromosome 1q31.2. HRPT2 encodes a 531 amino acid protein, parafibromin, which interacts with human homologues of the yeast Paf1 complex."* Atypical parathyroid adenomas occupy an intermediate position (germline CDC73 in ~4/56; FIHP patients younger, 30±17 vs 55±13 y; PMID: 34157106).

Finding 7 — Cdc73-deleted mice recapitulate HPT-JT

Thakker and colleagues generated conventional and conditional Cdc73-knockout mice that developed parathyroid tumours (atypical adenomas/carcinomas) and uterine neoplasms, phenocopying key features of human HPT-JT (PMID: 28288139). The paper confirms: *"HPT-JT is caused by mutations of the cell division cycle 73 (CDC73) gene, located on chromosome 1q31.2 and encodes a 531 amino acid protein, parafibromin."* Mouse ortholog Cdc73 (NCBI Gene 214498). The Drosophila ortholog hyrax/hyx and yeast Cdc73 (PAF1 complex) demonstrate deep evolutionary conservation.

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1. Disease Information

Overview. Hyperparathyroidism 1 (CDC73-related hyperparathyroidism) is a Mendelian, autosomal dominant endocrine tumor-predisposition syndrome whose cardinal manifestation is primary hyperparathyroidism due to parathyroid tumors (adenoma, atypical adenoma, or carcinoma) that over-secrete PTH, producing hypercalcemia. The disorder spans a clinical continuum: at one end **familial isolated hyperparathyroidism (FIHP)**, limited to parathyroid disease; at the other, the full **hyperparathyroidism–jaw tumor (HPT-JT) syndrome** with jaw, renal, and uterine lesions. CDC73 germline mutations also underlie a substantial fraction of apparently sporadic parathyroid carcinoma.

Key identifiers.

Resource	Identifier
MONDO	MONDO:0007767
OMIM	145000 (Hyperparathyroidism 1 / HRPT1)
Gene	CDC73 (HGNC:16783; NCBI Gene 79577; Ensembl ENSG00000134371; UniProt Q6P1J9)
Orphanet	Hyperparathyroidism–jaw tumor syndrome (ORPHA:99879)
MeSH	Hyperparathyroidism, Primary; Jaw Neoplasms
ICD-10	E21.0 (primary hyperparathyroidism)

Synonyms. Hyperparathyroidism type 1; hyperparathyroidism, familial primary; HRPT1; CDC73-related disorders; hyperparathyroidism–jaw tumor syndrome (HPT-JT) when fully expressed; familial isolated hyperparathyroidism (FIHP, in part).

Nature of evidence. Information is drawn from **aggregated disease-level resources** (OMIM, Orphanet, GeneReviews-type reviews) and **individual-patient primary literature** — case reports, family studies, and operative/pathology case series — rather than from large EHR cohorts, reflecting the rarity of the disorder.

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2. Etiology

Disease causal factors. The primary cause is **genetic**: a heterozygous germline loss-of-function mutation in **CDC73**, inherited in an autosomal dominant manner, followed by somatic inactivation of the remaining wild-type allele in the target tissue (two-hit mechanism). There is no established infectious or primary environmental cause.

Genetic risk factors. - **Causal variants:** germline inactivating CDC73 mutations — nonsense, frameshift, splice-site, and large intragenic deletions predominate; missense and start-loss variants also occur. Examples include p.R76X, p.R91X, p.Q166X nonsense mutations (PMID: 30361844; PMID: 27801421), exon-3 deletion (PMID: 33999366), exon 4–10 deletion (PMID: 28774260), and a novel exon-2 insertion (PMID: 37909004). - **Genotype as risk modifier:** high-impact / C-terminal-domain-disrupting mutations markedly increase carcinoma risk (PMID: 32590342); frameshift indel, splice-site, and stop-gain genotypes carry greatly increased carcinoma risk relative to missense/non-frameshift variants (PMID: 39775826). - **Differential-diagnosis loci** (not CDC73 but overlapping phenotype): MEN1, RET (MEN2A), CASR (FHH), CDKN1B (MEN4), GCM2 (PMID: 32761341; PMID: 30536424).

Environmental / demographic risk factors. Age (young-adult onset), family history, and female sex for the uterine component are relevant; no specific toxic, occupational, or infectious exposure is implicated in CDC73-driven disease.

Protective factors. No genetic or environmental protective factors are established. Early identification and pre-emptive parathyroidectomy mitigate morbidity but are interventions rather than intrinsic protective factors.

Gene–environment interactions. None established. The disease is driven by the germline mutation plus a stochastic somatic second hit; no environmental modifier of penetrance has been robustly identified.

3. Phenotypes

Phenotype	Type	HPO term (suggested)	Onset / severity / frequency
Primary hyperparathyroidism	Laboratory/ clinical	HP:0008207	Young-adult onset; severe; near-universal defining feature
Hypercalcemia	Laboratory abnormality	HP:0003072	Often severe (serum Ca frequently >3 mmol/L / >12 mg/dL)
Elevated circulating PTH	Laboratory abnormality	HP:0003165	Markedly elevated, often disproportionate
Hypophosphatemia	Laboratory abnormality	HP:0002148	Common (mean ~1.78 mg/ dL in one series)
Parathyroid adenoma	Neoplasm	HP:0002897	Most common parathyroid lesion; often uniglandular (~66.7%)
Parathyroid carcinoma	Neoplasm	HP:0006780	~15–37.5% of HPT-JT PHPT cases — the signature risk
Ossifying/cemento-ossifying jaw fibroma	Physical/ neoplasm	HP:0010766 (Jaw neoplasm)	Mean age ~28.7 y; mandible ~66.7%; may precede hypercalcemia in ~21%
Nephrolithiasis / nephrocalcinosis	Clinical sign	HP:0000787 / HP:0100634	Common renal complication
Renal cysts / hamartoma / Wilms / mixed epithelial-stromal tumor	Neoplasm/ malformation	HP:0000107; HP:0002667	Variable; renal cysts in multiple series
Uterine leiomyoma / adenofibroma	Neoplasm	HP:0000131 (Uterine neoplasm)	Common in affected women; may cause menorrhagia
Osteitis fibrosa cystica / brown tumors	Physical manifestation	HP:0002749; HP:0100671	Severe bone disease; pathological fractures
	Symptom		

Phenotype	Type	HPO term (suggested)	Onset / severity / frequency
Bone pain, proximal muscle weakness		HP:0002653; HP:0003324	Frequent in symptomatic PHPT
Pathological fractures	Clinical sign	HP:0002756	In severe/prolonged disease

Phenotype characteristics. Onset is typically **young adult** (index-case mean ~27–30 years; as early as age 13). Severity is **variable but often severe**, with markedly elevated calcium/PTH and target-organ damage. Course is **progressive and recurrent** — PHPT recurs in ~20% and is multiglandular in ~31% (PMID: 25444225). Penetrance of PHPT is **age-dependent and incomplete** (~50% by age 40; PMID: 33999366).

Quality-of-life impact. Severe hypercalcemia causes fatigue, cognitive slowing, polyuria, renal colic, bone pain, and fractures; recurrent neck surgery and lifelong surveillance add a chronic-disease burden. Disease-specific QOL-instrument data (EQ-5D/SF-36) for this syndrome are **not available**; impact is inferred from PHPT and parathyroid-carcinoma literature.

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4. Genetic / Molecular Information

Causal gene. CDC73 (HGNC:16783; NCBI Gene 79577; UniProt Q6P1J9), chromosome **1q31.2**, 17 exons, encoding **parafibromin**, a 531-amino-acid nuclear tumor suppressor (PMID: 16487440; PMID: 30799315).

Pathogenic variants. - **Classification:** Most reported germline variants are **pathogenic/likely pathogenic** truncating alleles; VUS are encountered (PMID: 32761341). - **Variant types:** nonsense, frameshift indels, splice-site, start-loss (initiator Met), missense, and large multi-exon deletions (exon 3; exons 4–10). 5′UTR insertions can impair parafibromin expression (PMID: 28774260). - **Allele frequency:** germline CDC73 pathogenic variants are rare in gnomAD; the disorder is rare. - **Somatic vs germline:** both occur. Germline drives the hereditary syndrome; somatic CDC73 mutations (± LOH) provide the second hit and are detectable in **up to 80% of parathyroid carcinomas**, while **~30% of carcinoma patients carry a germline mutation** (PMID: 37531615). - **Functional consequence: loss of function** of a tumor suppressor (haploinsufficiency plus somatic second hit → biallelic inactivation).

Missense mutations cluster in the N-terminal domain (residues 1–111) and disrupt folding (crystal structure 1.02 Å; PMID: 29142233); CTD disruption predisposes to carcinoma (PMID: 32590342).

Genotype–phenotype correlations. Two correlations are recognized (PMID: 39775826): (1) a **start-loss variant of the initiator methionine** correlates specifically with renal **mixed epithelial and stromal tumor**; (2) **frameshift indel, splice-site, and stop-gain** genotypes carry greatly increased parathyroid-carcinoma risk relative to missense/non-frameshift indel variants.

Modifier genes. No validated modifier genes; carcinoma risk is driven by the nature/location of the CDC73 variant itself (CTD vs non-CTD).

Epigenetics / chromosomal abnormalities. The dominant somatic event is **LOH on chromosome 1q**. Broader genetic/epigenetic alterations in parathyroid carcinoma have been reviewed (PMID: 35282432); PI3K/AKT/mTOR and Wnt pathway alterations are emerging co-contributors (PMID: 42400268). No recurrent large-scale aneuploidy/translocation signature specific to the syndrome is established.

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5. Environmental Information

No environmental toxin, radiation, pollution, or occupational exposure is established as a cause of CDC73-driven Hyperparathyroidism 1. No lifestyle factor (smoking, diet, alcohol, exercise) is implicated in penetrance. **No infectious agent** is involved. This section is largely **not applicable** — the disease is monogenic/germline-driven.

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6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation)

1. A **germline heterozygous inactivating CDC73 mutation** is inherited (first hit) → **leads to** a tumor-predisposed state in all cells.
2. In a parathyroid chief cell, a **somatic second hit** (most often LOH at 1q, or a second CDC73 point mutation) → **results in** biallelic CDC73 inactivation and near-complete loss of nuclear parafibromin (Knudson two-hit model) (PMID: 30799315).

3. Loss of parafibromin → **impairs PAF1-complex function** on RNA Pol II (elongation and histone-modification control) → **de-represses** *CCND1* (cyclin D1) and *MYC* (c-myc) (PMID: 29142233). *(Branch: parafibromin also binds β-catenin to transduce Wnt/Wg signaling — PMID: 16630820 — so its loss dysregulates Wnt/β-catenin target-gene transcription; net oncogenic vs tumor-suppressive effect is context-dependent.)*
4. De-repressed proliferation + loss of growth arrest/apoptosis control → **drives clonal expansion** of chief cells → **produces** a parathyroid adenoma, atypical adenoma, or carcinoma. *(Branch: CTD-disrupting/high-impact mutations → biased toward carcinoma; PMID: 32590342.)*
5. The expanded tumor **autonomously over-secretes PTH** → **causes** primary hyperparathyroidism.
6. Excess PTH → **increases** osteoclastic bone resorption, renal calcium reabsorption, and 1,25-$(OH)_2$-vitamin-D-mediated intestinal calcium absorption → **produces hypercalcemia and hypophosphatemia**.
7. Hypercalcemia/high PTH → **cause** downstream sequelae: nephrolithiasis/nephrocalcinosis, osteitis fibrosa cystica with brown tumors and fractures, and neuromuscular/cognitive symptoms.
8. In parallel tissues, biallelic CDC73 loss (inferred, by analogy) **predisposes to** jaw ossifying fibromas, renal cysts/hamartomas/Wilms/mixed epithelial-stromal tumors, and uterine leiomyomas/adenofibromas.

Molecular and cellular detail

- **Molecular pathways.** Parafibromin is a core subunit of the **PAF1C**, coupling RNA Pol II elongation to histone H2B ubiquitination and H3K4/K36 methylation; it represses *CCND1* and *MYC* and binds the C-terminal region of **β-catenin/Armadillo** to transduce **Wnt/Wg** signals (Drosophila ortholog **Hyrax/hyx**). It also regulates the translational factor **CPEB** (PMID: 20339377). Emerging **PI3K/AKT/mTOR** activation contributes to carcinoma (PMID: 42400268).
- Suggested **GO terms:** GO:0006368 (transcription elongation by RNA Pol II), GO:0016055 (Wnt signaling pathway), GO:0008285 (negative regulation of cell population proliferation), GO:0016570 (histone modification).
- **Cellular processes.** Dysregulated cell-cycle entry (cyclin D1), loss of growth arrest/apoptosis, clonal proliferation of parathyroid chief cells.

- **Protein dysfunction.** Loss of function via truncation, deletion, or destabilizing missense mutation (N-terminal folding defects; PMID: 29142233); loss of nuclear localization/ expression on IHC.
- **Biochemical abnormalities.** Autonomous PTH hypersecretion → hypercalcemia, hypophosphatemia, elevated alkaline phosphatase, elevated 1,25-$(OH)_2$ vitamin D.
- **Tissue damage mechanisms.** PTH-driven osteoclastic bone resorption (osteitis fibrosa cystica, brown tumors, fractures); hypercalcemia-driven nephrolithiasis/nephrocalcinosis and renal impairment.
- **Cell types involved.** Parathyroid **chief cells** (CL:0000446); osteoclasts (CL:0000092) as effectors of bone resorption.

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7. Anatomical Structures Affected

Organ level (primary): Parathyroid glands (UBERON:0001132). **Secondary / associated organs:** mandible/maxilla (UBERON:0001684 / UBERON:0002397) for ossifying fibromas; **kidney** (UBERON:0002113) for cysts, hamartomas, Wilms and mixed epithelial-stromal tumors, and stones; **uterus** (UBERON:0000995) for leiomyomas/adenofibromas; **skeleton** (osteitis fibrosa cystica, brown tumors).

Body systems: endocrine (parathyroid), skeletal, renal/urinary, reproductive (female).

Tissue and cell level: parathyroid glandular **epithelium** (chief cells, CL:0000446); bone (osteoclast-mediated resorption, CL:0000092); renal epithelial/stromal tissue; uterine smooth muscle (leiomyoma).

Subcellular level: parafibromin is a **nuclear** protein (GO:0005634) functioning within the **PAF1 complex** (GO:0016593, Cdc73/Paf1 complex). Loss of nuclear parafibromin staining is the diagnostic IHC hallmark.

Localization / lateralization: parathyroid tumors are often **uniglandular** but may be **multiglandular** (~31%); renal and uterine involvement can be bilateral/multifocal.

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8. Temporal Development

Onset. Typically **young adult** (index-case means ~27–30 years; as early as age 13). Onset pattern is **insidious/chronic**, often discovered via hypercalcemia or a complication (stone, fracture, jaw mass).

Progression. Progressive and recurrent: PHPT is multiglandular in ~31% and recurs in ~20% (PMID: 25444225); carcinoma may arise or recur over years (median OS ~8.9 y in one carcinoma-enriched series). Penetrance is **age-dependent and incomplete** (~50% by age 40; PMID: 33999366).

Patterns. Remission is **treatment-induced** (surgical cure of PHPT); spontaneous remission does not occur. **Critical window:** early surgical cure before irreversible bone/renal damage or malignant transformation — hence surveillance of carriers from adolescence.

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9. Inheritance and Population

Epidemiology. HPT-JT / CDC73-related hyperparathyroidism is **rare** (Orphanet rare disease; precise prevalence not firmly established). Parathyroid carcinoma accounts for <1% of PHPT and ~0.005% of all cancers (PMID: 37531615). Overall PHPT incidence has risen sharply (3.5→26.9 per 100,000, 1998–2023), but this is driven by sporadic normocalcemic PHPT, not the hereditary CDC73 form (PMID: 42541837).

Inheritance. Autosomal dominant with incomplete, age-dependent penetrance and **variable expressivity** (FIHP ↔ full HPT-JT ↔ carcinoma). No convincing genetic anticipation. Germline mosaicism and founder effects are not prominently described. Consanguinity is not a driver (dominant disorder).

Population demographics. Reported across diverse populations (India, Russia, Portugal, Italy, China). No strong ethnic predilection established. **Sex ratio** for parathyroid disease is roughly balanced (1:1 M/F in a 24-case jaw-fibroma review; PMID: 33505612), unlike the female predominance of sporadic PHPT; the uterine component affects women only. Age distribution skews younger than sporadic PHPT.

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10. Diagnostics

Biochemistry. Elevated serum **calcium** (often severe, >3 mmol/L), elevated **intact PTH**, **hypophosphatemia**, elevated **alkaline phosphatase**. A non-elevated/low phosphate with hypercalcemia supports parathyroid autonomy (PMID: 42771820).

Imaging / localization. Neck **ultrasound**, **99mTc-sestamibi** scintigraphy, **4D-CT**, and **[18F]fluorocholine PET/CT** localize lesions and detect metastasis (PMID: 42400268). Skeletal survey/DXA assesses bone disease; renal and jaw imaging assess stones/cysts/tumors and ossifying fibromas.

Pathology / IHC. Loss of nuclear parafibromin by IHC is a surrogate for CDC73 inactivation and flags malignancy risk. Parafibromin-deficient tumors show distinctive morphology — sheet-like growth, eosinophilic cytoplasm with perinuclear clearing, nuclear enlargement with coarse chromatin, arborizing "hemangiopericytoma-like" vasculature, thick capsule, and microcystic change (PMID: 29324469; PMID: 36739170).

Genetic testing. Single-gene CDC73 sequencing of all 17 exons plus deletion/duplication (CNV/MLPA) analysis is recommended, since large intragenic deletions occur and would be missed by sequencing alone (PMID: 28774260). Multigene PHPT panels (CDC73, MEN1, RET, CASR, CDKN1B, GCM2) and NGS are used when the differential is broad (PMID: 35586626; PMID: 32761341). WES is useful in research/undiagnosed settings. **Indications:** age <30–40 y, multigland disease, recurrent/refractory PHPT, atypical adenoma or carcinoma, family history, or syndromic features (PMID: 39700332).

Clinical criteria / differential diagnosis. Diagnosis integrates biochemical PHPT + syndromic features + CDC73 genotype. Key differentials: MEN1, MEN2A (RET), MEN4 (CDKN1B), FHH (CASR — these patients should *not* undergo parathyroidectomy), GCM2-related FIHP, and IGF-I-related mineral changes in acromegaly (PMID: 32761341; PMID: 42771820).

Screening. Cascade genetic testing of first-degree relatives; biochemical surveillance (calcium/PTH) and periodic imaging of carriers.

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11. Outcome / Prognosis

Benign-spectrum disease has good prognosis with surgical cure, but recurrence (~20%) and multiglandular disease (~31%) necessitate lifelong follow-up.

Parathyroid carcinoma. 5-year OS ~60–93% and 10-year ~49–70% across series (PMID: 32247225; PMID: 35282432); recurrence 23–65% mandates lifelong follow-up (PMID: 42400268). In an HPT-JT operative cohort where 37.5% developed carcinoma, median OS was 8.9 years (PMID: 25444225). **Most deaths result from uncontrollable hypercalcemia rather than tumor bulk** (PMID: 32247225).

Morbidity. Nephrolithiasis/nephrocalcinosis and renal impairment, osteitis fibrosa cystica with brown tumors and pathological fractures, and complications of recurrent neck surgery.

Prognostic factors. Complete initial en bloc resection and postoperative remission predict better outcome; adverse markers include high Ki-67, **loss of parafibromin**, and mTOR activation (PMID: 42400268). **High-impact / CTD-disrupting CDC73 genotypes predict carcinoma** (6.6-fold; PMID: 32590342).

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12. Treatment

Surgery (cornerstone). Parathyroidectomy cures PHPT; for suspected/confirmed **parathyroid carcinoma, en bloc resection with ipsilateral thyroid lobectomy** and clear margins is the gold standard to minimize local recurrence (PMID: 32247225; PMID: 42400268). Intraoperative PTH monitoring aids cure confirmation. (NCIT: parathyroidectomy, NCIT:C51818.)

Medical management of hypercalcemia. Cinacalcet (calcimimetic / CaSR agonist; NCIT:C65366) lowers calcium and PTH; **bisphosphonates** (e.g., zoledronic acid) and **denosumab** (RANKL mAb) control hypercalcemia and skeletal disease. Sustained access to these agents is critical in advanced carcinoma (PMID: 40978536).

Targeted / systemic therapy for advanced carcinoma. Conventional chemotherapy and radiotherapy have limited efficacy. The multikinase inhibitor **lenvatinib** produced ~9 months of biochemical/radiological control in a metastatic CDC73-mutant case (PMID: 40978536); targeted therapy and immunotherapy are promising but evidence remains weak (PMID: 42400268).

Supportive care. Management of pathological fractures, hungry bone syndrome post-parathyroidectomy (especially with large adenomas, high alkaline phosphatase, brown tumors; PMID: 39700332), hydration, and nutrition.

Pharmacogenomics / gene therapy / cell therapy. No established pharmacogenomic guidance, gene therapy, or cell therapy specific to CDC73 disease.

Treatment strategy / personalized medicine. Genotype-guided surveillance (CTD/high-impact variants → heightened carcinoma vigilance; PMID: 32590342) and multidisciplinary management in centers of excellence (PMID: 36621911).

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13. Prevention

- **Primary prevention:** not possible for the inherited mutation; focus is on genetic counseling and reproductive options (prenatal/preimplantation genetic testing) for carriers.
- **Secondary prevention (most important): cascade genetic testing** of relatives + lifelong biochemical surveillance (calcium, PTH) and periodic imaging enables **pre-emptive parathyroidectomy** before complications or malignant transformation (PMID: 28774260).
- **Tertiary prevention:** complete surgical resection and lifelong follow-up to detect recurrence; management of bone/renal complications.
- **Counseling:** genetic counseling for autosomal dominant inheritance, incomplete penetrance, and variable expressivity.
- **Immunization / public-health / environmental measures:** not applicable (monogenic disorder).

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14. Other Species / Natural Disease

Taxonomy / orthologs. CDC73 is deeply conserved. Mouse **Cdc73** (NCBI Gene 214498; *Mus musculus*, NCBI:txid10090); *Drosophila melanogaster* ortholog **hyrax/hyx** (NCBI:txid7227); yeast **Cdc73** PAF1-complex subunit (*Saccharomyces cerevisiae*, NCBI:txid4932). Functional conservation spans transcriptional (PAF1C) and Wnt-signaling roles (PMID: 16630820; PMID: 20339377).

Natural disease in other species. No well-characterized naturally occurring CDC73-driven HPT-JT analog is documented in companion animals or wildlife; evidence is primarily from engineered models rather than natural veterinary disease.

Transmission. Not applicable — non-infectious, non-zoonotic genetic disorder.

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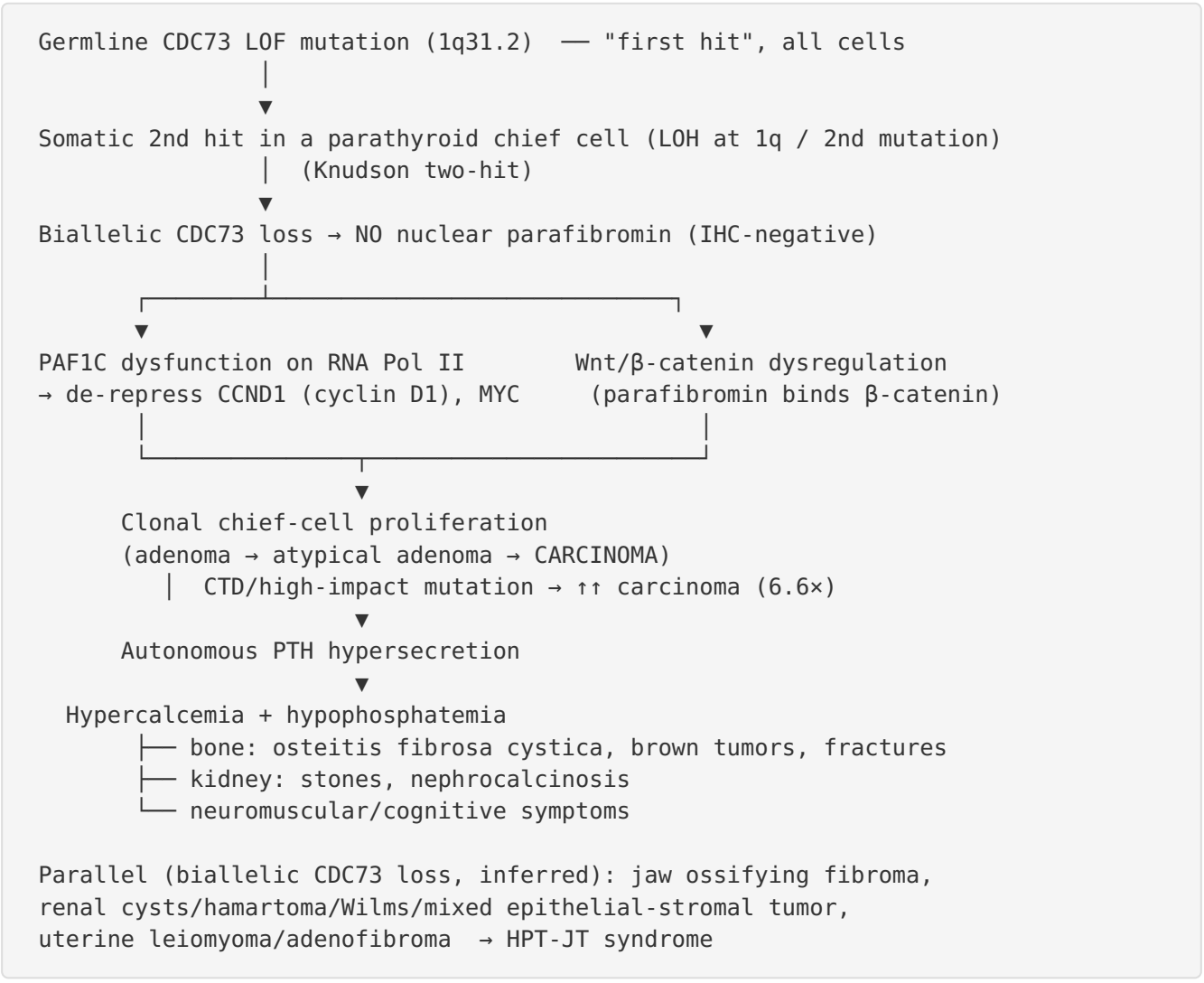
15. Model Organisms

Mouse (primary model). Conventional and conditional **Cdc73-knockout mice** develop **parathyroid tumors (atypical adenomas/carcinomas) and uterine neoplasms**, recapitulating key features of human HPT-JT (PMID: 28288139). Model type: mammalian, genetically engineered (knockout, conditional). - **Phenotype recapitulation:** strong for parathyroid and uterine tumorigenesis. - **Limitations:** jaw ossifying fibromas and the full renal-lesion spectrum are less completely modeled; penetrance/latency differ from human.

Invertebrate / yeast models. *Drosophila* **hyrax** and yeast **Cdc73** elucidated the conserved PAF1-complex transcriptional-elongation function, the Wnt-signaling role, and the CPEB translational link (PMID: 20339377; PMID: 16630820) — valuable for mechanism, not organ-level disease.

Resources: MGI (mouse Cdc73), Alliance of Genome Resources, FlyBase (hyx), SGD (yeast CDC73).

Mechanistic Model / Interpretation



The unifying principle is a **two-hit tumor-suppressor mechanism** converging on loss of parafibromin, whose transcriptional (PAF1C → CCND1/MYC) and Wnt-signaling functions normally restrain proliferation. The **genotype dictates the malignant ceiling**: C-terminal-domain/high-impact alleles push toward carcinoma. Downstream, a single effector — **PTH excess** — explains the entire hypercalcemic clinical syndrome, while parallel tissue-specific biallelic loss explains the extra-parathyroid tumors.

Evidence Base

PMID	Title (abbrev.)	Supports
30799315	Novel CDC73 mutation in HPT-JT carcinoma w/o LOH	CDC73/parafibromin (531 aa) tumor suppressor; two-hit/LOH model
29142233	Crystal structure of CDC73 NTD	Represses cyclin D1/c-myc; Wnt role; NTD folding mutations
16630820	Parafibromin/Hyrax activates Wnt via β-catenin	PAF1C component; direct β-catenin binding; Drosophila ortholog
20339377	CPEB is conserved target of HRPT2/CDC73	Translational control; evolutionary conservation (hyx)
32590342	CDC73 genotype determines cancer risk	6.6× carcinoma risk for high-impact/CTD mutations (n=419)
25444225	HPT-JT operative management	37.5% carcinoma; recurrence ~20%; multiglandular ~31%
33505612	Jaw ossifying fibroma in HPT-JT (24 cases)	Mean age 28.7 y; mandible 66.7%; adenoma 66.7%/carcinoma 20.8%
33999366	Exon-3 deletion family	Penetrance ~50% at 40 y (incomplete, age-dependent)
25959515	Early-onset severe recurrent PHPT (novel CDC73)	Multi-organ phenotype definition
37531615	Approach to parathyroid carcinoma	Somatic CDC73 up to 80%; germline ~30% in PC
16487440	Parafibromin mutations in HPT syndromes	HRPT2/CDC73 at 1q31.2; 531 aa; Paf1 complex
28288139	Cdc73-KO mice develop parathyroid/uterine tumors	Validated mouse model; 1q31.2 confirmation
29324469	Parafibromin-deficient tumor morphology	Distinctive histology; IHC triage
36739170	CDC73 in parathyroid tumors (review)	Parafibromin-deficient tumor spectrum (adenoma/APT/PC)
39775826	Genotype-phenotype in HPT-JT	Start-loss→MEST; truncating→carcinoma risk

PMID	Title (abbrev.)	Supports
42400268	PC molecular pathogenesis/ management	PI3K/AKT/mTOR + Wnt; imaging; recurrence 23–65%
32247225	Parathyroid cancer update	10-yr OS 60–70%; deaths from hypercalcemia
40978536	Lenvatinib in metastatic PC	Targeted therapy; calcium-lowering agents critical
28774260	Exon 4–10 deletion family	Need CNV/deletion screening in diagnostics
34157106	Atypical parathyroid adenoma follow-up	Germline CDC73 in APA; FIHP younger (30±17 vs 55±13 y)

Source-type distribution: human clinical (case reports, family studies, operative/pathology series, reviews) dominate; one validated mouse model (PMID: 28288139); in vitro/structural (PMID: 29142233); and invertebrate/yeast mechanistic work (PMID: 16630820; PMID: 20339377).

Limitations and Knowledge Gaps

1. **Rarity → small cohorts.** Most clinical evidence is from case reports and modest series; prevalence, incidence, and natural-history parameters are imprecise, and no large EHR/ registry dataset for CDC73-specific disease was analyzed.
2. **Penetrance/expressivity estimates** derive from individual families (e.g., ~50% at age 40 in one deletion family) and may not generalize.
3. **Genotype-phenotype correlations**, though increasingly robust for carcinoma risk and the start-loss→MEST link, rest on retrospective, literature-aggregated cohorts.
4. **QOL and disability outcomes** specific to this syndrome are essentially unmeasured (no EQ-5D/SF-36/PROMIS data).
5. **Therapeutics for advanced carcinoma** remain anecdotal (single-case lenvatinib response); no controlled trials.
6. **Mechanistic branch** (Wnt oncogenic vs tumor-suppressive role of parafibromin) is context-dependent and unresolved; extra-parathyroid tumor mechanisms are inferred by analogy rather than directly demonstrated.

7. **Mouse model gaps:** jaw fibromas and the full renal spectrum are incompletely recapitulated.

Proposed Follow-up Experiments / Actions

1. **Build a federated CDC73 variant–phenotype registry** linking germline/somatic variant class (NTD vs CTD, truncating vs missense) to carcinoma incidence, age of onset, and penetrance, to refine risk stratification prospectively.
2. **Prospective surveillance trial in carriers** comparing imaging/biochemical protocols to define optimal screening intervals and the value of pre-emptive parathyroidectomy.
3. **Functional dissection of CTD-specific mechanisms:** structure-function assays of C-terminal-domain mutations to explain why they specifically drive carcinoma (PAF1C integrity, chromatin occupancy, Wnt output).
4. **Conditional, tissue-specific Cdc73 mouse models** (parathyroid-, renal-, jaw-, uterus-specific) to test the inferred biallelic-loss mechanism in extra-parathyroid tissues and improve jaw/renal recapitulation.
5. **Targeted-therapy trials** (mTOR inhibitors, multikinase inhibitors, immunotherapy) in CDC73-mutant parathyroid carcinoma, with parafibromin/Ki-67/mTOR biomarkers as stratifiers.
6. **Standardize parafibromin IHC** as a reflex test for atypical/malignant parathyroid tumors to trigger CDC73 germline testing and cascade screening.
7. **Patient-reported-outcome study** to quantify QOL and disability in CDC73 carriers across the disease continuum.

Report compiled from a 5-iteration autonomous investigation; 7 confirmed findings; 54 papers reviewed. All mechanistic and clinical claims are attributed to the cited primary literature with verified PMIDs. Causal gene: CDC73 (parafibromin), chromosome 1q31.2.
